

Notes: Niehaus and Sukhtankar (AEJ: EP, 2013)

Corruption Dynamics: The Golden Goose Effect

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1 Big picture

- **Research question.** Do corrupt officials take less today when stealing more today raises the chance of losing valuable future opportunities to steal? More specifically, do expected future illicit rents deter current corruption in a large public employment program?
- **Main idea.** Standard efficiency-wage models emphasize licit future rents: officials avoid corruption today to keep their jobs and future wages. Niehaus and Sukhtankar emphasize that corrupt officials also value *illicit* future rents. This creates a “golden goose” effect: an official may avoid killing the goose that lays future golden eggs.
- **Empirical setting.** The paper studies India’s National Rural Employment Guarantee Scheme (NREGS), a public employment program that guarantees rural households up to 100 days of paid work per year.
- **Key shock.** On May 1, 2007, Orissa raised the statutory daily wage for unskilled NREGS workers from Rs 55 to Rs 70. This increased the value of fake daily-wage claims but did not directly change piece-rate project payments.
- **Corruption measure.** The paper compares official administrative microrecords to original household survey data. The gap between what officials reported and what households actually worked or received measures theft.
- **Main finding.** The wage increase had a static price effect that made daily-wage overreporting more attractive, but it also had a dynamic deterrent effect. Theft fell on piece-rate projects, and corruption fell differentially in villages with more future daily-wage projects.
- **Magnitude.** The authors estimate that golden goose effects reduced the increase in theft caused by the wage change by roughly 64 percent, though the confidence interval is wide.
- **Policy takeaway.** Anticorruption policy should account for dynamic incentives. A crackdown, pilot, wage increase, or audit may change not only today’s payoff from stealing, but also officials’ expectations of future rents.

2 Incremental contribution of the paper

- **From licit to illicit future rents.** Earlier corruption models focused on official wages, pensions, or wage premia as future rents that discipline agents. This paper shows that future *corrupt* rents can play the same disciplining role.
- **A clear dynamic mechanism.** The paper formalizes the trade-off between stealing more today and preserving future opportunities to steal. This makes static comparative statics potentially misleading.
- **Direct measurement of corruption.** Instead of using perceptions or audit flags, the paper measures corruption by matching official records to household survey reports at the work-spell and panchayat-day level.

- **A policy shock that separates static and dynamic effects.** The Orissa wage increase directly changed daily-wage corruption opportunities but not piece-rate opportunities. This allows the authors to isolate future-rent effects using piece-rate projects.
- **Forward-looking tests.** The paper constructs measures of future project composition, especially the share of upcoming projects that are daily wage. This tests whether officials react to future rent opportunities rather than only current opportunities.
- **Evidence against simple substitution stories.** By comparing future project composition with past project composition, the authors distinguish a forward-looking golden goose effect from time-symmetric mechanisms such as income targeting or mechanical intertemporal substitution.
- **Implications for program evaluation.** A temporary pilot can generate very different incentives from permanent implementation. Pilots may understate or overstate corruption depending on how they affect future rent expectations.

3 Context, data, and measurement

3.1 NREGS background

NREGS is a large rural employment program in India. During the study period, every rural household was legally entitled to apply for up to 100 days of paid work per year. Projects were planned by local governments, approved higher in the administrative hierarchy, and implemented at the panchayat level.

- **Projects.** Typical projects include road construction, irrigation, water conservation, and earthworks.
- **Payment types.** Workers can be paid either a statutory daily wage or a piece rate, depending on whether output is easy to measure.
- **Officials.** Implementation involves appointed bureaucrats and local elected officials, including block officials, panchayat secretaries, field assistants, mates, and the elected sarpanch.
- **Monitoring.** Official guidelines call for local social audits and top-down inspections, but in the study setting systematic audits were weak and punishment probabilities were positive but small.

Interpretation. The program creates repeated opportunities for local officials to control reporting, payments, and documentation. This makes it a natural setting for studying dynamic corruption incentives.

3.2 Forms of theft

The paper focuses on theft from the labor budget, which is measured by comparing official claims to household survey reports.

- **Underpayment.** A worker performs real work but receives less than the amount officially owed.
- **Overreporting.** Officials report more work or payments than actually occurred, including fictitious work spells or inflated days.
- **Total theft.** For a daily-wage project, if the statutory wage is w^s , the worker receives w^a , actual days are n , and reported days are $\hat{n}$, then rents are

$$(w^s - w^a)n + (\hat{n} - n)w^s.$$

The first term is underpayment; the second term is overreporting.

Interpretation. Workers have some incentive to notice and complain about underpayment, but less incentive to monitor overreporting because the program budget is not fixed at the village level.

3.3 The Orissa wage shock

The empirical design uses a statutory wage increase in Orissa. The daily wage for unskilled NREGS work rose from Rs 55 to Rs 70, announced on April 28, 2007 and effective May 1, 2007.

- The shock directly increased the amount officials could steal per fictitious day on daily-wage projects.
- The shock did not directly change piece-rate project payments during the study window.
- The shock was specific to Orissa, so the authors use neighboring Andhra Pradesh as a control for some piece-rate specifications.
- The project shelf was largely predetermined before the wage change, which helps make future project composition plausibly exogenous to the shock.

Interpretation. Daily-wage projects identify the net effect of static and dynamic forces; piece-rate projects are especially useful because they should respond only through future-rent expectations.

3.4 Official data and survey data

The authors use public official NREGS records and original household survey data.

- **Official data.** Administrative records report worker names, addresses, caste status, household jobcard numbers, dates and duration of work, projects, and payments.

- **Survey data.** Households were asked about actual NREGS work between March 1 and June 30, 2007, including dates, days worked, projects, payment type, and amounts received.
- **Study area.** The main Orissa sample comes from Gajapati, Koraput, and Rayagada districts near the Andhra Pradesh border. Vizianagaram district in Andhra Pradesh is used as a control in some analyses.
- **Sampling.** The authors sampled officially recorded work spells and then surveyed the alleged beneficiary households. The original Orissa sample contained 1,938 households; 1,328 household interviews were completed, and 821 households confirmed NREGS participation during the study period.
- **Aggregation.** Official and survey reports are aggregated to panchayat-day outcomes, allowing the authors to estimate changes around the May 1 wage shock.

Interpretation. The data are unusually direct: corruption is measured as discrepancies between what the state paid out according to official records and what households say they actually worked or received.

3.5 Main variables

- **Official daily-wage days.** Number of daily-wage work days officially reported at the panchayat-day level.
- **Actual daily-wage days.** Number of daily-wage days reported by surveyed households.
- **Official piece-rate payments.** Total official payments on piece-rate projects at the panchayat-day level.
- **Actual piece-rate payments.** Total piece-rate payments reported by surveyed households.
- *FwdWageFrac.* Fraction of project-days in the next two months that are daily-wage project-days.
- *FdwAll* and *FdwSome.* Indicators for whether all, or some but not all, upcoming project-days are daily wage.
- *BdwAll* and *BdwSome.* Analogous indicators using past project-days, used to test time-symmetric substitution mechanisms.

Interpretation. The most important constructed variable is future project composition. It proxies for how valuable future corruption opportunities become after the wage increase.

4 Methodology and empirical specifications

4.1 Dynamic rent-extraction model

The model has an official who can overreport work but faces a probability of detection and removal. The official's continuation value includes both legal compensation and future opportunities to steal.

Let ϕ be the probability that a future project is daily wage. The official's continuation value can be written as

$$V(w^s, \phi) = \phi V^W(w^s, \phi) + (1 - \phi) V^P(w^s, \phi),$$

where V^W is the value in a daily-wage period and V^P is the value in a piece-rate period.

For daily-wage projects, the official chooses reported days $\hat{n}$:

$$V^W(w^s, \phi) = \max_{\hat{n}} \left[(w^s - w^a)n + (\hat{n} - n)w^s + \beta(1 - \pi(n, \hat{n}))V(w^s, \phi) \right].$$

For piece-rate projects, the analogous problem is

$$V^P(w^s, \phi) = \max_{\hat{q}} \left[(r^s - r^a)q + (\hat{q} - q)r^s + \beta(1 - \mu(q, \hat{q}))V(w^s, \phi) \right].$$

- Higher overreporting raises current rents.
- Higher overreporting also raises the chance of detection and job loss.
- A permanent wage increase raises the value of future daily-wage corruption opportunities.
- This higher continuation value makes current detection more costly, so officials may steal less today.

4.2 Main theoretical predictions

- **Proposition 1: daily-wage overreporting is ambiguous.** A permanent daily-wage increase has a static price effect that encourages overreporting and a golden goose effect that discourages overreporting. The net effect depends on the elasticity of future rents with respect to the wage.
- **Proposition 2: piece-rate theft should fall.** Piece-rate projects are not directly affected by the daily-wage increase, so the main predicted effect is dynamic: officials steal less to preserve future rent opportunities.
- **Proposition 3: stronger effects when more future projects are daily wage.** The golden goose effect should be stronger in panchayats with more daily-wage projects coming up, because the wage hike increases future rents more in those places.

Interpretation. Static models predict more theft when the price of theft rises. The dynamic model predicts that some of that increase is offset, and on margins not directly affected by the price change theft should fall.

4.3 Baseline shock specification

The first empirical specification relates official reports to actual reports and the wage shock:

$$\hat{y}_{pt} = \beta_0 + \beta_1 y_{pt} + \beta_2 Shock_t + T_t' \gamma + R_p' \zeta + \delta_{d(p)} + \varepsilon_{pt}.$$

Here $\hat{y}_{pt}$ is the official outcome, y_{pt} is the actual survey outcome, $Shock_t$ equals one on and after May 1, T_t contains time controls, R_p contains political reservation controls, and $\delta_{d(p)}$ are district fixed effects.

- For daily-wage outcomes, $\hat{y}_{pt}$ is officially reported days.
- For piece-rate outcomes, $\hat{y}_{pt}$ is officially reported payments.
- Standard errors are clustered by both panchayat and day.

4.4 Difference-in-differences with Andhra Pradesh

For piece-rate theft, the authors also use Andhra Pradesh as a control:

$$\begin{aligned} \hat{y}_{pt} = & \beta_0 + \beta_1 y_{pt} + \beta_2 (ORShock_t \times OR_p) + \beta_3 (APShock1_t \times AP_p) \\ & + \beta_4 (APShock2_t \times AP_p) + \beta_5 ORShock_t + \beta_6 APShock1_t + \beta_7 APShock2_t \\ & + OR_p + T_t' \gamma + R_p' \zeta + \delta_{d(p)} + \varepsilon_{pt}. \end{aligned}$$

The coefficient of interest is β_2 , the Orissa-specific post-shock change in piece-rate reporting.

4.5 Future project composition

The key heterogeneity specification allows the shock to have different effects depending on the future share of daily-wage projects:

$$\begin{aligned} \hat{y}_{pt} = & \beta_0 + \beta_1 y_{pt} + \beta_2 Shock_t + \beta_3 (Shock_t \times FdwAll_{pt}) + \beta_4 FdwAll_{pt} \\ & + \beta_5 (Shock_t \times FdwSome_{pt}) + \beta_6 FdwSome_{pt} + T_t' \gamma + R_p' \zeta + \delta_{d(p)} + \varepsilon_{pt}. \end{aligned}$$

- For daily-wage reports, the static price effect suggests $\beta_2 > 0$, while the golden goose effect suggests $\beta_3 < 0$.
- For piece-rate reports, the direct price effect is absent, so the shock and its future-project interactions should be negative if golden goose effects dominate.

4.6 Forward-looking versus time-symmetric mechanisms

Alternative substitution mechanisms can also reduce current theft when other theft opportunities rise. However, many of these mechanisms are time-symmetric: past and future rent opportunities should matter similarly. The authors therefore estimate specifications that include both future and backward-looking project-composition variables.

- If the golden goose effect is correct, future daily-wage opportunities should matter more than past opportunities.
- If income targeting, limited corruption inputs, or mechanical time aggregation drive the result, past and future opportunities should have more symmetric effects.

4.7 Monitoring tests

A separate concern is that monitoring intensity changed around the wage shock. The authors use village-level data on the most recent visits by Block Development Officers and District Collectors, estimating maximum-likelihood models of visit probabilities over time.

Interpretation. The monitoring tests find no evidence that the May 1 wage shock increased official visits. This weakens the view that the main results are driven by higher audit intensity rather than future-rent incentives.

5 Identification and interpretation

5.1 What identifies the golden goose effect

The central identification strategy combines three sources of variation.

- **Time variation:** corruption is observed before and after the May 1 wage increase.
- **Project-type variation:** daily-wage projects are directly affected by the wage increase; piece-rate projects are not directly affected.
- **Future-composition variation:** panchayats differ in how many upcoming projects are daily wage, changing the size of the future-rent effect.

Interpretation. The strongest test is piece-rate theft: if a daily-wage increase reduces corruption on piece-rate projects, the channel is unlikely to be a current price effect.

5.2 Why the shock is useful

The wage increase was set by policymakers above the local officials being studied. It was also specific to Orissa, and it did not immediately update piece-rate schedules. This makes it a useful shock to the value of future daily-wage rents.

- A temporary daily-wage increase would mainly raise the current price of overreporting.
- A permanent daily-wage increase also raises future rent expectations.
- The difference between temporary and permanent effects is the dynamic golden goose force.

5.3 Project composition concern

A key concern is that officials might endogenously switch toward daily-wage projects after the wage increase. The paper addresses this by showing that *FwdWageFrac*, the future daily-wage fraction, does not jump after the shock.

Interpretation. This supports the claim that project composition reflects a predetermined project shelf rather than manipulation in response to the wage change.

5.4 Survey measurement concern

The corruption measure depends on household recall and truthful reporting. The authors reduce this concern by surveying a recent period, using reference points to help recall, interviewing households independently of the government, and focusing on differential changes around the policy shock.

Interpretation. Level measurement error could affect the estimated amount of corruption, but the main tests rely on changes and interactions around the shock.

5.5 Alternative substitution mechanisms

The paper discusses three alternatives: scarce inputs to produce corruption, income effects or income targeting, and time aggregation in monitoring. These mechanisms can make current theft fall when other theft opportunities rise.

Key distinction. These mechanisms tend to be time-symmetric: past and future opportunities should have similar effects. The data show stronger and more consistent relationships with future opportunities, supporting a forward-looking interpretation.

5.6 Monitoring concern

Monitoring by senior officials could confound the results if it increased exactly where future daily-wage opportunities were high. The visit data do not support this explanation.

Interpretation. The evidence does not show an increase in monitoring after May 1, and the official wage-change notification did not include new monitoring provisions.

5.7 Interpreting the magnitude

The authors' rough calculations imply that golden goose effects reduced the increase in theft caused by the wage hike by about 64 percent. They also estimate that dynamic effects reduced daily-wage overreporting by 49 percent and piece-rate theft by 77 percent.

Caution. These magnitudes should be read carefully because some coefficients are imprecise and the confidence interval around the 64 percent estimate is wide.

6 Tables and figures

6.1 Figure 1 and Figure 2: Project-type distributions

Read:

- Figure 1 shows that projects are usually close to pure daily-wage or pure piece-rate projects, rather than mixtures of both payment types.
- Figure 2 shows the distribution of future daily-wage project shares. There are strong mass points at zero and one.
- These figures justify using project-type variation and future project composition as key empirical objects.

Economic message: The empirical design relies on identifiable daily-wage and piece-rate projects, and on meaningful cross-panchayat variation in upcoming project types.



FIGURE 1. DISTRIBUTION OF PROJECT TYPES

Notes: Plots distribution of projects in study panchayats by the fraction of spells of (reported) work done that were daily wage spells. Work spells are coded as daily wage spells if the payment per day is one of the statutory daily wages. (Orissa implements four different daily wages for varying skill levels).

Figure 1: Distribution of project types.

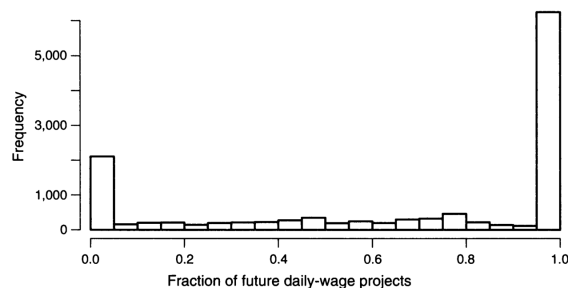


FIGURE 2. DISTRIBUTION OF FUTURE DAILY-WAGE PROJECT FRACTION

Note: Plots distribution of projects in study panchayats by the fraction of projects in the subsequent two months that were daily-wage projects.

Figure 2: Distribution of future daily-wage project fraction.

6.2 Figure 3 and Table 1: Wage implementation and core summary statistics

Read Figure 3:

- Official daily-wage claims rise sharply after the May 1 shock, showing that the statutory wage increase was implemented in administrative records.
- Actual wages received by workers do not rise in the same way.
- The visual pattern implies that the wage increase mainly changed official claims rather than worker take-home pay.

Read Table 1:

- Official daily-wage days average 3.31 per panchayat-day, while actual daily-wage days average 0.88.

- Official piece-rate payments average Rs 94.08, while actual piece-rate payments average Rs 12.96.
- The mean future daily-wage fraction is 0.67, so future daily-wage opportunities are common.

Economic message: The administrative data show much larger work and payment claims than the household survey data, indicating substantial leakage.

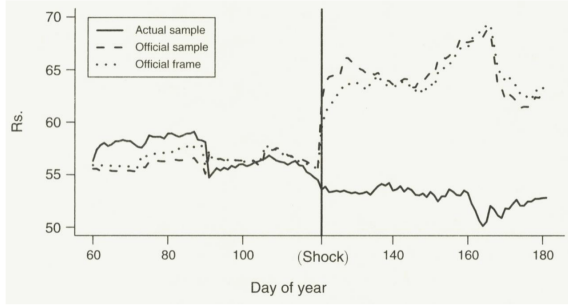


FIGURE 3. DAILY-WAGE RATES PAID

Notes: Plots a daily series of the average wage rate paid in daily-wage projects in Orissa over the study period, according to official records and survey data. Day 60 corresponds to March 1, 2007, the start of the study period; day 121 to May 1, 2007, the date of the wage shock; and day 181 to June 30, 2007, the end of the study period.

Figure 3: Daily-wage rates paid.

TABLE 1—SUMMARY STATISTICS OF MAIN REGRESSION VARIABLES

	Observations	Mean	SD
Official DW days	13,054	3.31	6.30
Actual DW days	13,054	0.88	1.55
Official PR payments	7,320	94.08	259.70
Actual PR payments	7,320	12.96	43.43
<i>FwdWageFrac</i>	13,908	0.67	0.40

Notes: This table provides summary descriptions of the aggregated variables used in the main result Tables 3 and 4. The sample for each kind of project includes panchayats that had at least one of that kind of project active during the study period (March 1 through June 30, 2007). “Official DW Days” is the days worked by panchayat-day on daily wage projects as reported officially. “Actual DW Days” is the days worked by panchayat-day on daily wage projects as reported by survey respondents. “Official PR Rate” is the total payments by panchayat-day on piece-rate projects as reported officially, while “Actual PR Rate” corresponds to the same figure as reported by survey respondents. “FwdWageFrac” is the proportion of project-days in the next two months in a panchayat that are daily wage.

Table 1: Summary statistics of main regression variables.

6.3 Table 2 and Figure 4: Project composition and daily-wage overreporting

Read Table 2:

- The dependent variable is the fraction of future project-days that are daily wage.
- The shock coefficients are small and statistically insignificant.
- This supports the assumption that future project composition did not endogenously change after the wage increase.

Read Figure 4:

- The figure plots overreporting of daily-wage work days around the May 1 wage shock.
- The fitted discontinuous polynomial models do not show a large unconditional jump in daily-wage overreporting.
- This muted response is consistent with the static price effect being offset by a dynamic golden goose effect.

Economic message: The wage shock did not reshape the project shelf, and the daily-wage corruption response is not simply the large static increase a one-period model would predict.

TABLE 2—WAGE SHOCK EFFECTS ON PROJECT COMPOSITION

Regressor	(1)	(2)	(3)
<i>Shock</i>	0.014 (0.021)	0.007 (0.019)	0.008 (0.018)
<i>Day</i>	0.001 (0.001)	0.001 (0.001)	−0.003 (0.002)
<i>Day</i> ²			0.002 (0.001)
District FEs	N	Y	Y
Observations	12,103	12,103	12,103
<i>R</i> ²	0.046	0.097	0.098

Notes: Each observation is a panchayat-day. The dependent variable in all regressions is “*FwdWageFrac*,” the proportion of daily wage project-days in the panchayat in the next two months. “*Shock*” is an indicator equal to 1 on and after May 1, 2007. “*Day*” is a linear time trend; *Day*² has been rescaled by the mean of *Day*. All columns include a third-order polynomial in the day of the month and indicators for major agricultural seasons. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 2: Wage shock effects on project composition.

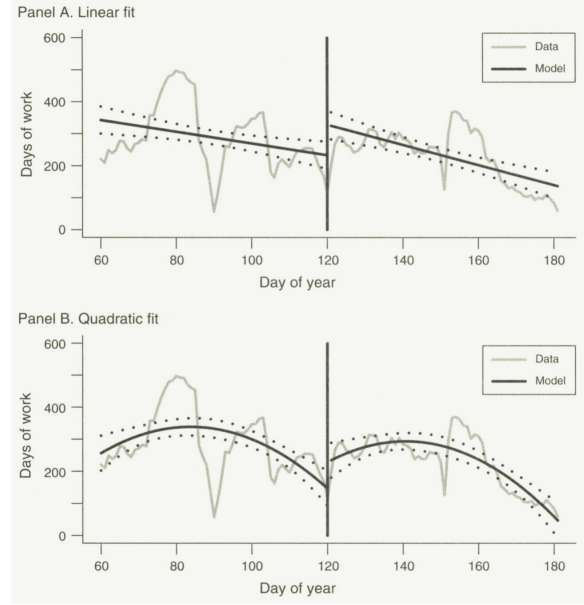


FIGURE 4. DAILY WAGE CORRUPTION MEASURES WITH DISCONTINUOUS POLYNOMIAL FITS

Notes: Plots daily series of the total amount of overreporting of work days on daily wage projects. Day 60 corresponds to March 1, 2007, the start of the study period; day 121 to May 1, 2007, the date of the wage shock; and day 181 to June 30, 2007, the end of the study period. Discrepancies were calculated by subtracting the quantities reported by survey respondents from those reported in official records. Superimposed solid lines represent fitted regression discontinuity models with linear (panel A) and quadratic (panel B) terms in day-of-year; dotted lines represent 95 percent confidence intervals.

Figure 4: Daily-wage corruption measures with discontinuous polynomial fits.

6.4 Tables 3 and 4: Main daily-wage and piece-rate regressions

Read Table 3:

- Panel A shows that the overall daily-wage shock effect is positive but not always statistically significant.
- Panchayats with only daily-wage projects experience a differential reduction in overreporting.
- Panel B shows that the shock increases daily-wage reports in places with no future daily-wage projects, but the interaction with *FdwAll* is negative.
- The future interaction is stronger when backward-looking controls are included, consistent with forward-looking incentives.

Read Table 4:

- Piece-rate reports fall after the wage shock, even though piece-rate payments were not directly affected by the daily-wage increase.
- The main shock coefficient in Panel A is around -78 to -82 rupees and statistically significant in most specifications.

- Dynamic interaction estimates are negative but less precise, reflecting limited power for piece-rate observations.

Economic message: Table 3 shows the offset between static and dynamic forces for daily-wage reports. Table 4 provides a cleaner golden goose test because piece-rate theft falls on a margin not directly affected by the wage hike.

TABLE 3—WAGE SHOCK EFFECTS ON DAILY-WAGE REPORTS

Regressor	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Wage shock effects</i>						
<i>Shock</i>	0.95 (0.78)	0.94 (0.78)	0.89 (0.78)	1.30* (0.79)	1.29 (0.79)	1.24 (0.80)
<i>Shock × AlwaysDW</i>				-1.75* (1.00)	-1.74* (0.98)	-1.75* (0.99)
<i>AlwaysDW</i>				2.12** (0.83)	2.27*** (0.86)	2.28*** (0.86)
Observations	12,810	12,810	12,810	12,810	12,810	12,810
R ²	0.08	0.09	0.09	0.09	0.10	0.10
<i>Panel B. Wage shock dynamic effects</i>						
<i>Shock</i>	2.39** (0.95)	2.31** (0.96)	2.25** (0.95)	3.05** (1.22)	3.00** (1.23)	3.00** (1.23)
<i>Shock × FdwAll</i>	-1.94* (1.07)	-1.84* (1.07)	-1.80* (1.07)	-4.03*** (1.38)	-3.78*** (1.36)	-3.78*** (1.37)
<i>Shock × FdwSome</i>	-1.15 (1.03)	-1.12 (1.03)	-1.08 (1.02)	-0.21 (0.94)	-0.17 (0.94)	-0.17 (0.94)
<i>Shock × BdwAll</i>				2.27 (1.50)	2.13 (1.46)	2.12 (1.47)
<i>Shock × BdwSome</i>				-1.99** (0.94)	-2.03** (0.97)	-2.03** (0.97)
Observations	11,386	11,386	11,386	10,651	10,651	10,651
R ²	0.09	0.09	0.09	0.13	0.14	0.14
Time controls	<i>Day</i>	<i>Day</i>	<i>Shock × Day</i>	<i>Day</i>	<i>Day</i>	<i>Shock × Day</i>
District FEs	N	Y	Y	N	Y	Y

Notes: Each observation is a panchayat-day. The dependent variable in all regressions is the number of days of daily-wage work officially reported. “Shock” is an indicator equal to one on and after May 1, 2007; in columns 3 and 6, it is the intercept difference at the time the shock occurs. “AlwaysDW” is a panchayat that had a daily-wage project active throughout the study period. “FdwAll” is equal to one if the proportion of daily wage project-days in the panchayat in the next two months is equal to 1, and “BdwAll” is the analogous variable for the preceding two months. “FdwSome” is equal to 1 if the proportion of daily wage project-days in the next two months is greater than 0 but less than 1, and “BdwSome” is the analogous variable for the preceding two months. All regressions include controls for the number of days of daily-wage work reported by participants, an indicator for major holidays, a third-order polynomial in the day of the month, indicators for major agricultural seasons, and indicators for the panchayat chief seat being reserved for a minority group. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 3: Wage shock effects on daily-wage reports.

TABLE 4—WAGE SHOCK EFFECTS ON PIECE-RATE REPORTS

Regressor	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Wage shock effects</i>						
<i>Shock</i>	-78.31** (39.91)	-78.43* (40.29)	-75.9* (40.08)	-81.76** (40.26)	-82.18** (40.66)	-79.87** (40.58)
<i>Shock × AlwaysPR</i>				15.44 (50.43)	16.64 (49.80)	17.58 (49.36)
<i>AlwaysPR</i>				-35.29 (33.87)	-33.19 (34.83)	-33.58 (34.73)
Observations	7,076	7,076	7,076	7,076	7,076	7,076
R ²	0.04	0.05	0.05	0.04	0.05	0.05
<i>Panel B. Wage shock dynamic effects</i>						
<i>Shock</i>	-38.58 (67.50)	-40.47 (66.52)	-38.18 (67.18)	-63.69 (73.19)	-62.16 (72.35)	-60.53 (72.34)
<i>Shock × FdwAll</i>	-24.88 (69.39)	-20.36 (67.39)	-23.75 (68.79)	-44.14 (93.40)	-31.83 (90.06)	-39.19 (93.11)
<i>Shock × FdwSome</i>	-74.61 (72.18)	-73.94 (69.87)	-72.84 (69.81)	-74.85 (95.70)	-73.83 (94.34)	-73.46 (94.20)
<i>Shock × BdwAll</i>				109.23 (81.61)	105.72 (81.84)	113.68 (84.81)
<i>Shock × BdwSome</i>				11.94 (89.23)	5.17 (89.35)	8.55 (90.37)
Observations	6,543	6,543	6,543	6,209	6,209	6,209
R ²	0.08	0.08	0.08	0.11	0.11	0.12
Time controls	<i>Day</i>	<i>Day</i>	<i>Shock × Day</i>	<i>Day</i>	<i>Day</i>	<i>Shock × Day</i>
District FEs	N	Y	Y	N	Y	Y

Notes: Each observation is a panchayat-day. The dependent variable in all regressions is the total amount paid on piece-rate projects officially reported. “Shock” is an indicator equal to 1 on and after May 1, 2007; in columns 3 and 6, it is the intercept difference at the time the shock occurs. “AlwaysPR” is a panchayat that had a piece rate project active throughout the study period. “FdwAll” is equal to one if the proportion of daily wage project-days in the panchayat in the next two months is equal to 1, and “BdwAll” is the analogous variable for the preceding two months. “FdwSome” is equal to 1 if the proportion of daily wage project-days in the next two months is greater than 0 but less than 1, and “BdwSome” is the analogous variable for the preceding two months. All regressions include controls for the number of days of daily-wage work reported by participants, an indicator for major holidays, a third-order polynomial in the day of the month, indicators for major agricultural seasons, and indicators for the panchayat chief seat being reserved for a minority group. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 4: Wage shock effects on piece-rate reports.

6.5 Figure 5 and Table 5: Piece-rate evidence and Andhra Pradesh control

Read Figure 5:

- Piece-rate theft falls sharply around the May 1 wage shock.
- This decline is visible in both linear and quadratic fitted models.
- Piece-rate theft later rebounds, which is consistent with changing future rent expectations as the season approaches the monsoon slowdown.

Read Table 5:

- The table uses Andhra Pradesh as a control for piece-rate reports.
- The Orissa-specific shock effect is negative, around -88 rupees, and statistically significant.

- This strengthens the interpretation that the Orissa wage shock reduced piece-rate theft through a dynamic channel.

Economic message: The piece-rate results are the most direct evidence of golden goose effects because they show a reduction in theft where the wage shock did not directly increase current returns.

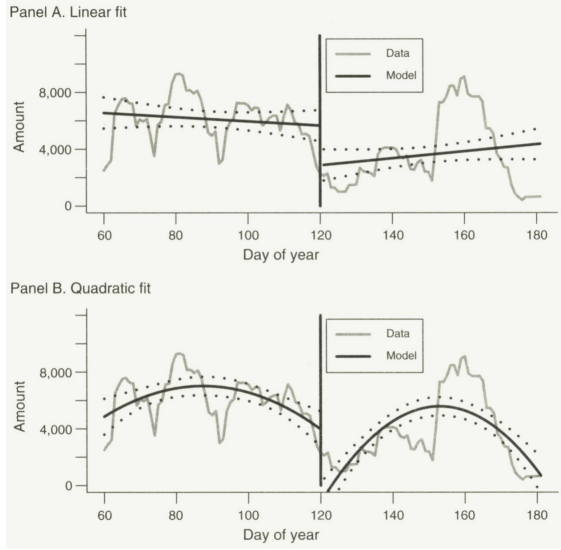


FIGURE 5. PIECE RATE CORRUPTION MEASURES WITH DISCONTINUOUS POLYNOMIAL FITS

Notes: Plots daily series of the total amount of theft on piece rate projects. Day 60 corresponds to March 1, 2007, the start of the study period; day 121 to May 1, 2007, the date of the wage shock; and day 181 to June 30, 2007, the end of the study period. Discrepancies were calculated by subtracting the quantities reported by survey respondents from those reported in official records. Superimposed solid lines represent fitted regression discontinuity models with linear (panel A) and quadratic (panel B) terms in day-of-year; dotted lines represent 95 percent confidence intervals.

Figure 5: Piece-rate corruption measures with discontinuous polynomial fits.

TABLE 5—EFFECTS ON PIECE-RATE REPORTS USING ANDHRA PRADESH AS A CONTROL

Regressor	(1)	(2)	(3)
<i>OR Shock</i> × <i>OR</i>	−87.86** (38.81)	−87.90** (38.77)	−87.54** (38.86)
<i>AP Shock 1</i> × <i>AP</i>	−21.29 (30.09)	−21.45 (29.99)	−21.03 (30.14)
<i>AP Shock 2</i> × <i>AP</i>	117.84*** (33.87)	117.95*** (33.83)	119.38*** (34.05)
<i>OR Shock</i>	31.15 (32.51)	31.40 (32.38)	53.64 (32.88)
<i>AP Shock 1</i>	61.08** (27.42)	60.69** (27.50)	23.38 (25.78)
<i>AP Shock 2</i>	−24.34 (25.89)	−24.71 (25.85)	−63.81** (26.00)
Actual PR payments	0.19** (0.08)	0.19** (0.08)	0.19** (0.08)
Time controls	Day	Day	Shock × Day
FEs	State	District	District
Observations	16,470	16,470	16,470
<i>R</i> ²	0.06	0.06	0.06

Notes: This table uses data from both Orissa (OR) and Andhra Pradesh (AP). Each observation is a panchayat-day. The dependent variable in all regressions is the total amount paid out on piece-rate projects as officially reported. “*OR Shock*” is an indicator equal to 1 on and after May 1, 2007; in column 3, it is the intercept difference at the time the shock occurs. “*AP Shock 1*” is an indicator equal to one on and after March 5, 2007, while “*AP Shock 2*” equals one on or after April 25, 2007. All columns include a third-order polynomial in the day of the month, an indicator for major holidays, and indicators for major agricultural seasons. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 5: Piece-rate reports using Andhra Pradesh as a control.

6.6 Tables 6 and 7: Robustness and additional outcomes

Read Table 6:

- The main results are similar using one-month and two-month future windows.
- Three-month windows are weaker, which is plausible because NREGS activity declines with the monsoon.
- Month-specific specifications suggest some anticipatory adjustment may have begun before May.

Read Table 7:

- The results are similar when using total extraction from daily-wage and piece-rate projects.
- Results for fictitious households provide additional support for the daily-wage findings.

- The negative $Shock \times FdwAll$ terms remain central to the interpretation.

Economic message: The core patterns are not driven by one narrow outcome definition or one exact window for measuring future project composition.

TABLE 6—ROBUSTNESS CHECKS

Regressor	Daily wage				Piece rate			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Shock</i>	2.03** (0.89)	1.69** (0.84)		2.44** (0.97)	-69.45 (65.13)	-26.96 (75.21)	-37.90 (67.03)	
<i>Shock</i> $\times$ <i>FdwAll</i>	-1.75* (0.99)	-1.35 (1.04)		-1.94* (1.09)	1.98 (63.27)	-44.15 (77.92)	-22.86 (68.91)	
<i>Shock</i> $\times$ <i>FdwSome</i>	-1.28 (0.92)	-0.33 (0.98)		-1.19 (1.04)	-51.87 (64.72)	-96.82 (80.02)	-81.61 (72.31)	
April			0.29 (1.25)				103.47 (67.50)	
May			2.76 (1.77)				42.39 (76.83)	
June			3.37 (2.24)				205.76 (154.60)	
April $\times$ <i>FdwAll</i>			-0.35 (1.41)				-125.72** (57.67)	
May $\times$ <i>FdwAll</i>			-1.75 (1.71)				-39.77 (50.69)	
June $\times$ <i>FdwAll</i>			-2.60 (1.92)				-167.24 (119.43)	
Time window (months)	1	3	2	2	1	3	2	2
Observations	10,740	11,740	11,386	11,386	6,250	6,653	6,543	6,543
R^2	0.11	0.09	0.09	0.04	0.09	0.08	0.09	0.08

Notes: Each observation is a panchayat-day. The dependent variable in columns 1–3 is the number of days of daily-wage work officially reported; in column 4, the difference between this quantity and the number of days of daily-wage work reported by participants; in columns 5–7, the total amount paid out on piece-rate projects as officially reported; and in column 8, the difference between this quantity and the total amount paid out on piece-rate projects as reported by participants. “*Shock*” is an indicator equal to one on and after May 1, 2007. “*FdwAll*” is equal to one if the proportion of daily wage project-days in the panchayat in the next two months is equal to 1. “*FdwSome*” is equal to 1 if the proportion of daily wage project-days in the next two months is greater than 0 but less than 1. All columns include controls for the actual quantities of work done/amounts received, a linear time trend, an indicator for major holidays, a third-order polynomial in the day of the month, indicators for major agricultural seasons, and indicators for the panchayat chief seat being reserved for a minority group except columns 3 and 7 which omit the polynomial in day-of-month. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 7—ADDITIONAL OUTCOME VARIABLES

Regressor	DW+PR	Daily wage				Piece rate	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Shock</i>	105.40 (82.45)	138.55 (97.14)	174.64** (77.29)	1.21 (1.17)	0.79 (0.84)	-25.30 (39.99)	-34.97 (48.18)
<i>Shock</i> $\times$ <i>FdwAll</i>	-73.79 (94.20)	-69.49* (38.02)	-192.76*** (36.61)	-1.78** (0.74)	-0.88*** (0.33)	7.89 (80.37)	-5.96 (76.89)
<i>Shock</i> $\times$ <i>FdwSome</i>	-95.45 (95.29)	-39.40 (36.79)	22.35 (69.19)	0.08 (0.21)	-0.26*** (0.03)	3.76 (53.23)	-16.41 (36.14)
Time controls	Day	Day	Shock $\times$ Day	Day	Day	Day	Day
Fixed effects	District	District	District	District	District	District	District
Observations	12,103	11,386	10,651	9,885	10,433	5,614	5,828
R^2	0.04	0.04	0.08	0.05	0.03	0.08	0.06

Notes: Each observation is a panchayat-day. The dependent variable in column 1 is total extraction from daily-wage and piece-rate projects. In columns 2–3 it is the total value extracted from daily-wage projects. In columns 4–7 it is the number of daily-wage work done or piece-rate amounts for “fictitious” households as officially reported. “*Shock*” is an indicator equal to 1 on and after May 1, 2007. “*FdwAll*” is equal to 1 if the proportion of daily wage project-days in the panchayat in the next two months is equal to 1, and “*FdwSome*” is equal to 1 if the proportion of daily wage project-days in the next two months is greater than 0 but less than 1. All columns include controls for the actual quantities of work done/amounts received, a linear time trend, an indicator for major holidays, a third-order polynomial in the day of the month, indicators for major agricultural seasons, and indicators for the panchayat chief seat being reserved for a minority group. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 7: Additional outcome variables.

Table 6: Robustness checks.

6.7 Table 8 and Appendix Table B1: Monitoring and sample coverage

Read Table 8:

- The table estimates whether visits by Block Development Officers or District Collectors changed after the wage shock.
- There is no robust evidence of increased monitoring after May 1.
- This weakens the monitoring-confound explanation.

Read Appendix Table B1:

- The reached sample is similar to the initial sample on age, gender, post-shock status, spell length, wage-spell status, and daily rate.
- The main significant difference is SC/ST status, reflecting survey access constraints in tribal areas.
- Importantly, reachability does not differ meaningfully before versus after the wage change.

Economic message: The main results are less likely to be explained by changing monitoring intensity or by differential sample attrition around the wage shock.

TABLE 8—ML ESTIMATES OF CHANGING AUDIT PROBABILITIES OVER TIME

Regressor	BDO	BDO	Collector	Collector
<i>Shock</i>	0.049 (0.304)	0.07 (0.322)	0.105 (0.482)	−1.597 (0.753)**
<i>Koraput</i>	−3.007 (0.179)***	−2.996 (0.187)***	−4.769 (0.276)***	−4.854 (0.274)***
<i>Gajapati</i>	−4.771 (0.242)***	−4.761 (0.246)***	−5.742 (0.39)***	−5.83 (0.389)***
<i>Rayagada</i>	−3.872 (0.168)***	−3.862 (0.174)***	−5.425 (0.284)***	−5.51 (0.283)***
<i>Day</i>	0.082 (0.017)***	0.082 (0.018)***	0.048 (0.024)*	0.147 (0.038)***
<i>Day</i> ²		0 (0.001)		0.007 (0.002)***

Notes: This table presents maximum likelihood estimates of the probability of a visit by government officials—Block Development Officers (BDO) and District Collectors—to the panchayat. “*Shock*” is an indicator equal to 1 on and after May 1, 2007. “*t*” and “*t*²” are time trends. Koraput, Rayagada, and Gajapati are indicators for the three study districts in Orissa.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 8: ML estimates of changing audit probabilities over time.

TABLE B1—CHARACTERISTICS OF SPELLS IN SAMPLE FRAME, INITIAL SAMPLE, AND REACHED SAMPLE

Variable	All spells		Sampled spells		Reached spells		<i>p</i> -value
	Mean	SD	Mean	SD	Mean	SD	
Age	37.60	14.93	37.37	13.60	37.55	13.28	0.33
Male	0.54	0.50	0.54	0.50	0.54	0.50	0.67
SC/ST	0.78	0.41	0.79	0.41	0.77	0.42	0.05
Post	0.40	0.49	0.43	0.49	0.42	0.49	0.57
Spell length	11.13	2.92	11.14	3.01	11.09	3.14	0.33
Wage spell	0.83	0.37	0.83	0.38	0.84	0.36	0.20
Daily rate	63.48	17.24	64.37	20.34	63.90	18.92	0.30

Notes: Reports summary statistics at the work-spell level using official records and for (i) the universe of spells sampled from ($N = 111,172$), (ii) the initial sample of work spells we drew ($N = 7,126$), and (iii) the work spells done by households we were ultimately able to interview ($N = 4,794$). The last column reports the *p*-value from a regression of the variable in question on an indicator for whether or not the observation is in our analysis sample (conditional on being in our initial sample), with standard errors clustered at the panchayat level.

Appendix Table B1: Characteristics of spells in the sample frame, initial sample, and reached sample.

6.8 Appendix Tables B2 and C1: Survey sample and dependent-variable construction

Read Appendix Table B2:

- The surveyed households are poor and have low education levels.
- NREGS participants and nonparticipants look broadly similar on demographics and awareness measures.
- Awareness of formal rights, such as the Right to Information Act, is very low.

Read Appendix Table C1:

- The table gives a numerical example of how work spells are spread across days.
- This explains how household and official work-spell records become panchayat-day variables.

Economic message: The appendix clarifies how the analysis sample is constructed and how the core panchayat-day outcomes are built.

TABLE B2—SAMPLE DESCRIPTION

Variable	NREGA participants			Nonparticipants		
	Observations	Mean	SD	Observations	Mean	SD
Demographics						
Number of HH members	812	4.94	1.88	498	4.65	2.18
BPL card holder	815	0.77	0.42	497	0.76	0.43
HH head is literate	803	0.3	0.46	501	0.23	0.42
HH head educated through grade 10	819	0.04	0.19	502	0.04	0.2
Awareness						
Knows HH keeps job card	806	0.84	0.37	476	0.89	0.31
Number of amenities aware of	810	0.96	0.85	494	0.78	0.82
HH head has heard of RTI Act	821	0.02	0.13	501	0.01	0.09

Notes: This table describes attributes of the household survey sample that was successfully interviewed in Orissa. The sample is split between households who confirm that they worked on an NREGA project between March 1 and June 30, 2007—821 households (NREGA participants)—and those that did not (507 households). “BPL” stands for Below the Poverty Line, a designation that entitles one to several government programs, although makes no difference for NREGA work. The definition for literacy used by the Indian government is whether one can sign her name (instead of placing a thumbprint). The amenities meant to be provided at the worksite in NREGA projects are—amongst others—water, shade, first aid, and a creche/child care. We ask respondents to name amenities without prompting. “RTI” stands for the Right to Information Act, a freedom of information act passed by the Indian government in 2005.

C. Additional Appendix Tables

TABLE C1—NUMERICAL EXAMPLE OF DEPENDENT VARIABLE CONSTRUCTION

Worker	Report	Attributed work by day					
		May 1	May 2	May 3	May 4	May 5	May 6
A	3 days between May 1–6	0.5	0.5	0.5	0.5	0.5	0.5
B	4 days from May 3–6	0	0	1	1	1	1
Totals:		0.5	0.5	1.5	1.5	1.5	1.5

Notes: This table presents numerical examples of how our dependent variables were aggregated up to the panchayat-day level from official and survey reports of work done. The rows show two typical reports of work done within a panchayat; the columns show how we attributed the number of days reported as worked across the period during which they were worked, and summed them up for each panchayat-day record.

Appendix Table C1: Numerical example of dependent-variable construction.

Appendix Table B2: Sample description.

6.9 Appendix Tables C2 and C3: Quadratic time-trend robustness

Read:

- Appendix Table C2 repeats the daily-wage regressions with quadratic time trends.
- Appendix Table C3 repeats the piece-rate regressions with quadratic time trends.
- The qualitative patterns remain similar to the main tables.

Economic message: The main estimates are not an artifact of imposing only linear time controls around the wage shock.

Regressor	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Wage shock effects</i>						
<i>Shock</i>	0.88 (0.78)	0.88 (0.79)	1.04 (0.98)	1.23 (0.79)	1.23 (0.80)	1.40 (0.94)
<i>Shock × AlwaysDW</i>				−1.75* (1.01)	−1.75* (0.99)	−1.73* (0.99)
<i>AlwaysDW</i>				2.14** (0.84)	2.28*** (0.86)	2.27*** (0.86)
Observations	12,810	12,810	12,810	12,810	12,810	12,810
R ²	0.08	0.09	0.09	0.09	0.10	0.10
<i>Panel B. Wage shock dynamic effects</i>						
<i>Shock</i>	2.28** (0.94)	2.22** (0.95)	2.28* (1.26)	3.01** (1.24)	2.97** (1.24)	2.97* (1.53)
<i>Shock × FdwAll</i>	−1.83* (1.07)	−1.76* (1.06)	−1.78* (1.06)	−3.90*** (1.40)	−3.71*** (1.38)	−3.70*** (1.35)
<i>Shock × FdwSome</i>	−1.07 (1.02)	−1.05 (1.02)	−1.03 (1.02)	−0.17 (0.94)	−0.15 (0.94)	−0.11 (0.94)
<i>Shock × BdwAll</i>				2.17 (1.51)	2.07 (1.47)	2.10 (1.44)
<i>Shock × BdwSome</i>				−2.01** (0.95)	−2.04** (0.98)	−2.01** (0.94)
Observations	11,386	11,386	11,386	10,651	10,651	10,651
R ²	0.09	0.10	0.10	0.13	0.14	0.14
Time controls	Day2	Day2	Shock×Day2	Day2	Day2	Shock×Day2
District FEs	N	Y	Y	N	Y	Y

Notes: Each observation is a panchayat-day. The dependent variable in all regressions is the number of days of daily-wage work officially reported. “Shock” is an indicator equal to 1 on and after May 1, 2007; in columns 3 and 6, it is the intercept difference at the time the shock occurs. “AlwaysDW” is a panchayat that had a daily-wage project active throughout the study period. “FdwAll” is equal to 1 if the proportion of daily wage project-days in the panchayat in the next two months is equal to 1, and “BdwAll” is the analogous variable for the preceding two months. “FdwSome” is equal to 1 if the proportion of daily wage project-days in the next two months is greater than 0 but less than 1, and “BdwSome” is the analogous variable for the preceding two months. All regressions include controls for the number of days of daily-wage work reported by participants, an indicator for major holidays, a third-order polynomial in the day of the month, indicators for major agricultural seasons, and indicators for the panchayat chief seat being reserved for a minority group. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

Appendix Table C2: Daily-wage reports with quadratic time trends.

Regressor	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Wage shock effects</i>						
<i>Shock</i>	−78.02* (40.02)	−77.69* (40.25)	−107.05* (59.55)	−81.48** (40.38)	−81.52** (40.67)	−111.41** (56.18)
<i>Shock × AlwaysPR</i>				15.56 (50.35)	16.98 (49.60)	18.41 (48.96)
<i>AlwaysPR</i>				−35.38 (33.82)	−33.32 (34.78)	−34.25 (34.62)
Observations	7,076	7,076	7,076	7,076	7,076	7,076
R ²	0.04	0.05	0.05	0.04	0.05	0.06
<i>Panel B. Wage shock dynamic effects</i>						
<i>Shock</i>	−37.46 (67.85)	−39.62 (66.82)	−83.01 (73.64)	−63.16 (73.25)	−61.93 (72.47)	−100.15 (84.32)
<i>Shock × FdwAll</i>	−27.71 (70.79)	−22.54 (68.47)	−20.67 (67.62)	−50.13 (96.55)	−37.42 (92.82)	−36.27 (91.82)
<i>Shock × FdwSome</i>	−74.57 (72.20)	−73.74 (69.9)	−69.65 (69.30)	−75.65 (96.15)	−74.54 (94.64)	−69.08 (93.13)
<i>Shock × BdwAll</i>				114.07 (84.33)	111.48 (84.62)	115.15 (84.54)
<i>Shock × BdwSome</i>				14.69 (90.81)	8.19 (90.69)	4.83 (89.31)
Observations	6,543	6,543	6,543	6,209	6,209	6,209
R ²	0.08	0.08	0.09	0.11	0.12	0.12
Time controls	Day2	Day2	Shock×Day2	Day2	Day2	Shock×Day2
District FEs	N	Y	Y	N	Y	Y

Notes: Each observation is a panchayat-day. The dependent variable in all regressions is the total amount paid on piece-rate projects officially reported. “Shock” is an indicator equal to 1 on and after May 1, 2007; in columns 3 and 6, it is the intercept difference at the time the shock occurs. “AlwaysPR” is a panchayat that had a piece rate project active throughout the study period. “FdwAll” is equal to 1 if the proportion of daily wage project-days in the panchayat in the next two months is equal to 1, and “BdwAll” is the analogous variable for the preceding two months. “FdwSome” is equal to 1 if the proportion of daily wage project-days in the next two months is greater than 0 but less than 1, and “BdwSome” is the analogous variable for the preceding two months. All regressions include controls for the number of days of daily-wage work reported by participants, an indicator for major holidays, a third-order polynomial in the day of the month, indicators for major agricultural seasons, and indicators for the panchayat chief seat being reserved for a minority group. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

Appendix Table C3: Piece-rate reports with quadratic time trends.

6.10 Appendix Table C4: Andhra Pradesh control with quadratic time trends

Read:

- Appendix Table C4 repeats the Andhra Pradesh control specification with quadratic time trends.
- The Orissa-specific shock effect remains negative and statistically significant.

Economic message: The piece-rate difference-in-differences result survives a more flexible time-trend specification.

TABLE C4—EFFECTS ON PIECE-RATE REPORTS
USING ANDHRA PRADESH AS A CONTROL, QUADRATIC TIME TRENDS

Regressor	(1)	(2)	(3)
<i>OR Shock</i> × OR	−87.39** (38.94)	−87.31** (38.92)	−86.87** (38.93)
<i>AP Shock 1</i> × AP	−21.10 (30.29)	−21.24 (30.18)	−23.30 (30.18)
<i>AP Shock 2</i> × AP	119.97*** (34.13)	120.03*** (34.10)	119.74*** (34.08)
<i>OR Shock</i>	52.21 (32.00)	52.51 (31.94)	−35.07 (43.97)
<i>AP Shock 1</i>	−3.47 (26.52)	−3.57 (26.43)	18.89 (22.00)
<i>AP Shock 2</i>	−63.85*** (24.27)	−63.91*** (24.22)	−44.79 (27.81)
Actual PR payments	0.20** (0.08)	0.20** (0.08)	0.20** (0.08)
Time controls	Day2	Day2	Shock × Day2
FEs	State	District	District
Observations	16,470	16,470	16,470
<i>R</i> ²	0.06	0.06	0.07

Notes: This table uses data from both Orissa (OR) and Andhra Pradesh (AP). Each observation is a panchayat-day. The dependent variable in all regressions is the total amount paid out on piece-rate projects as officially reported. “*OR Shock*” is an indicator equal to 1 on and after May 1, 2007; in column 3, it is the intercept difference at the time the shock occurs. “*AP Shock 1*” is an indicator equal to 1 on and after March 5, 2007, while “*AP Shock 2*” equals 1 on or after April 25, 2007. All columns include a third-order polynomial in the day of the month, an indicator for major holidays, and indicators for major agricultural seasons. Robust standard errors multi-way clustered by panchayat and day are presented in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Appendix Table C4: Piece-rate reports using Andhra Pradesh as a control, quadratic time trends.

7 Short summary

- The paper studies whether expected future illicit rents discipline current corruption.
- The empirical setting is NREGS, India’s large rural employment guarantee program.
- Corruption is measured directly by comparing official records to household survey reports.
- The key shock is Orissa’s May 1, 2007 daily-wage increase from Rs 55 to Rs 70.
- A daily-wage increase has a static price effect: fake daily-wage days become more lucrative.
- A permanent daily-wage increase also has a dynamic golden goose effect: future rent opportunities become more valuable, so officials have more to lose from being caught.
- The paper uses piece-rate projects to isolate dynamic effects because the wage shock did not directly change piece-rate payments.
- Theft on piece-rate projects falls after the wage shock, both in Orissa-only specifications and relative to Andhra Pradesh.

- Daily-wage overreporting rises less where future daily-wage projects are more common.
- The future-composition effects are stronger than backward-looking effects, supporting a forward-looking interpretation.
- The authors find no evidence that monitoring visits increased after the wage shock.
- Rough calculations imply that golden goose effects reduced the impact of the wage increase on theft by about 64 percent.

8 Conclusion

8.1 Core conclusion

Niehaus and Sukhtankar show that corrupt officials respond to expected future illicit rents. When future stealing opportunities become more valuable, officials may steal less today in order to reduce the risk of losing access to those future rents.

8.2 Economic intuition

A permanent increase in the statutory daily wage makes fake daily-wage claims more profitable. A static model predicts more overreporting. But if officials expect many future daily-wage projects, the same wage increase also raises the value of keeping their position. This increases the cost of detection and can reduce current theft.

8.3 Incremental contribution revisited

The paper’s contribution is to make illicit future rents central to the study of corruption dynamics. It provides a theoretical framework, a direct measurement strategy, and empirical evidence from a large public program that future rents can discipline corrupt agents even when those future rents are themselves corrupt.

8.4 Implications

The results imply that policy design should consider how interventions change future rent expectations. Audits, wage changes, and program reforms can have displacement effects across corruption margins. Temporary pilots may not predict permanent implementation because they create different dynamic incentives.

8.5 Limitations

The setting is one program in two Indian states during a short period. Some estimates are imprecise, especially for piece-rate heterogeneity. The magnitude calculations are rough and

should not be treated as externally valid structural parameters. The survey-based measure of actual work and payments may contain recall or reporting error, though the identification relies mainly on differential changes around the shock.

8.6 Takeaway

The enduring takeaway is that corrupt officials are forward-looking. Future illicit rents can act like an efficiency wage: they make job loss costly and can restrain current theft. Ignoring this golden goose effect can lead to misleading predictions about the effects of anticorruption policy.